

Program : Diploma in Chemical Engineering	
Course Code : 6071C	Course Title : Energy Engineering
Semester : 6	Credits : 4
Course Category : Program Elective	
Periods per week : 4 (L: 4 T: 0 P: 0)	Periods per semester : 60

Course Objectives:

- To impart basic knowledge in the energy generation, storage and energy management.

Course Prerequisites:

Topic	Course code	Course name	Semester
General chemistry		Applied Chemistry	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Outline the forms of energy, its classification and conventional energy resources	12	Understanding
CO2	Summarize the nonconventional energy resources and power generation	18	Understanding
CO3	Summarize various energy storage technologies	18	Understanding
CO4	Summarize the energy management system.	12	Understanding

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1					3		
CO2					3		
CO3					3		
CO4					3		

3-Strongly Mapped 2-Moderately Mapped 1- Weakly Mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Outline the forms of energy, its classification and conventional energy resources		
M1.01	Summarise the concept of energy and its resources	1	Understanding
M1.02	Illustrate coal and its processing	4	Understanding
M1.03	Summarise oil, its exploration and processing	4	Understanding.
M1.04	Summarise nuclear energy and its reactors	3	Understanding
Contents: Energy-Different forms of energy-classification of energy resources- Conventional energy resources- wood- coal -oil and natural gas- nuclear energy Coal and its classification-Coal gasification- fluidized bed combustion-coal liquefaction-proximate and ultimate analysis-calorific value Oil- sources- composition- exploration- processing block diagram(upstream, midstream and downstream)- natural gas- sources- properties and composition Nuclear energy-nuclear fission –fissile elements-moderation and breeding- moderators and coolants -typical power reactor –pressurized water reactor-breeder reactor-pool reactor.			
CO2	Summarize the nonconventional energy resources and power generation		
M2.01	Summarise the nonconventional resources	1	Remembering
M2.02	Summarize the solar and biomass power generation	4	Understanding

M2.03	Summarize the geothermal energy power generation	3	Understanding
M2.04	Summarize the wind energy power generation	3	Understanding
M2.05	Summarize the ocean energy power generation	3	Understanding
M2.06	Summarize the hydro electric power generation	3	Understanding
	Series Test – I	1	

Contents:

Non-conventional energy resources-solar-biogas and biomass-ocean and tidal energy-geothermal-wind

Solar radiation-solar collector-solar trough systems-solar power towers-solar dish/engine system-solar thermal power plant-solar pond-solar water heater- solar cooker

Biogas and biomass-availability -biomass conversion processes- biogas generation-biogas production plants(batch ,continuous, movable drum type, fixed dome type)

Geothermal energy-resources of geothermal energy- vapor dominated power plant-liquid dominated system-total flow geothermal plant-merits and demerits of geothermal power generation- applications of geothermal energy.

Wind energy-wind turbine- wind power- power and velocity duration characteristics of wind-airfoil construction and types-types of wind power plants-types of wind genitor units with its construction-site selection-merits and demerits of wind power generation.

Ocean thermal energy conversion (OTEC)- working principle- types of OTEC systems-advantages and limitation. Ocean wave energy and tidal energy- tidal power plants.

Hydraulic energy-hydel power- principle of operation- hydro electric power systems

CO3	Summarize various energy storage technologies		
M3.01	Identifying different energy storage devices	2	Remembering.
M3.02	Summarize the chemical and electro chemical storage devices	4	Understanding
M3.03	Summarize the solar energy storage devices	4	Understanding
M3.04	Summarize fuel cell technology	4	Understanding
M 3.05	Summarize mechanical storage	4	Understanding

Contents:

Energy storage - storage types- electrochemical storage, mechanical storage, chemical storage, thermal storage

Chemical and electrochemical storage-components of battery systems-battery- lead acid battery-lithium ion battery- lithium ion capacitors

Renewable energy storage-solar cells- types of solar cells-solar cell power plant-solar photovoltaic cells-photo voltaic systems- applications of photo voltaic systems- advantages and limitations of solar energy systems

fuel cell- basic electro chemistry-component of polymer electrolyte membrane fuel cell-types of polymer electrolyte membrane- hydrogen production techniques- hydrogen storage; Composite tanks, Cryogenic liquid hydrogen (LH₂),Chemical hydrides, Carbon based materials, Metal hydrides- fuel cell system

Mechanical storage- pump hydroelectric storage-compressed air storage-flywheel-electromagnetic energy storage-thermal energy storage-sensible heat storage-water storage-packed bed exchange storage-latent heat energy storage

CO4	Summarize the energy management system.		
M4.01	Summarize the energy management systems	4	Understanding
M4.02	Summarize the energy audit	3	Understanding
M4.03	Summarise energy crisis and its management	2	Understanding
M4.04	Illustrate integrated energy systems	2	Understanding
	Series Test – II	1	

Contents:

Energy management system-principles of energy management- energy distribution management-energy – energy monitoring and its elements

Energy audit-objectives of audit-methodology

Energy crisis-causes-crisis management- energy crisis in India

Integrated energy systems –block diagram-types of integrated systems- biomass diesel integrated system- diesel photo voltaic integrated system

Text / Reference

T/R	Book Title/Author
T ₁	Non-conventional energy resources; S Hasan Saeed, D K Sharma,
R ₁	Energy Technology; S Rao Dr B.B. Parulekar

Online Resources

S.No	Website Link
1	https://nptel.ac.in/courses/103/103/103103206/